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Lyapunov–Schmidt Reduction for Fractional Differential Systems

Lyapunov–Schmidt reduction is one of the powerful and significant methods to simplify nonlinear ordinary differential systems (ODSs) with standard formulation. In this paper, we extend Lyapunov–Schmidt reduction to fractional ordinary differential systems (FODSs) with Caputo derivatives. [DOI: 10.1115/1.4033607]

Keywords: Lyapunov–Schmidt reduction, fractional differential equation, Caputo derivative, Fredholm operator

1 Introduction

Generally speaking, there are at least two reduction methods for ODEs, namely, center manifold reduction [1,2] and Lyapunov–Schmidt reduction [3–8]. By using these two reduction methods, the complex ODEs can be reduced to the relatively simple ones.

In recent decades, the fractional calculus has attracted increasing interest, in part because it is a good candidate to describe the dynamical phenomena possessing the memory effect compared to the ordinary differential system [9]. Recently, Ma and Li [10] extended the center manifold reduction to the FODSs which makes the analysis simplified. Is it possible to generalize Lyapunov–Schmidt reduction to FODSs? In this paper, we give an active answer.

The remainder of the paper is organized as follows: In Sec. 2, we briefly present the basic theorems of Lyapunov–Schmidt reduction which are used for ODEs. Section 3 presents Lyapunov–Schmidt method for FODSs. The last section concludes this paper.

2 Mathematical Preliminaries

In this section, we introduce some basic definitions and properties from fractional calculus and Lyapunov–Schmidt reduction which will be used later on.

Definition 1. *The left and right Riemann–Liouville fractional integrals of function $f(t)$ with order $\alpha > 0$ are defined as* [11,12]

$$ {}_{RL}D^{-\alpha}_{0,t}f(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-\tau)^{\alpha-1}f(\tau)\mathrm{d}\tau, \quad t > 0 \tag{1}$$

and

$$ {}_{RL}D^{-\alpha}_{t,b}f(t) = \frac{1}{\Gamma(\alpha)}\int_t^b (\tau-t)^{\alpha-1}f(\tau)\mathrm{d}\tau, \quad t < b \tag{2}$$

respectively, where $\Gamma(\cdot)$ is the gamma function.

Definition 2. *The left and right Riemann–Liouville fractional derivatives of function $f(t)$ with order α are defined as* [11,12]

$$ {}_{RL}D^{\alpha}_{0,t}f(t) = \frac{1}{\Gamma(n-\alpha)}\frac{d^n}{dt^n}\int_0^t (t-\tau)^{n-\alpha-1}f(\tau)\mathrm{d}\tau, \quad t > 0 \tag{3}$$

and

$$ {}_{RL}D^{\alpha}_{t,b}f(t) = \frac{(-1)^n}{\Gamma(n-\alpha)}\frac{d^n}{dt^n}\int_t^b (\tau-t)^{n-\alpha-1}f(\tau)\mathrm{d}\tau, \quad t < b \tag{4}$$

respectively, where $n-1 < \alpha < n \in Z^+$.

Definition 3. *The left and right Caputo derivatives of function $f(t)$ with order α are defined as* [11,12]

$$ {}_{C}D^{\alpha}_{0,t}f(t) = \frac{1}{\Gamma(n-\alpha)}\int_0^t (t-\tau)^{n-\alpha-1}f^{(n)}(\tau)\mathrm{d}\tau, \quad t \geq 0 \tag{5}$$

and

$$ {}_{C}D^{\alpha}_{t,b}f(t) = \frac{(-1)^n}{\Gamma(n-\alpha)}\int_t^b (\tau-t)^{n-\alpha-1}f^{(n)}(\tau)\mathrm{d}\tau, \quad t \leq b \tag{6}$$

respectively, where $n-1 < \alpha < n \in Z^+$.

It is known that Riemann–Liouville derivative and Caputo derivative are generally not equivalent to each other. But they have important relations between them. The following property can be known by simple calculations or referring to Ref. [13].

Property 1. For given $f(t)$ and $g(t)$, if $f(t)\cdot {}_{C}D^{\alpha}_{0,t}g(t)$ and $g(t)\cdot {}_{RL}D^{\alpha}_{t,b}f(t)$ are both locally integrable on $[0, b]$, then

$$\int_0^b f(t)\cdot {}_{C}D^{\alpha}_{0,t}g(t)\mathrm{d}t = \int_0^b g(t)\cdot {}_{RL}D^{\alpha}_{t,b}f(t)\mathrm{d}t + \sum_{j=0}^{n-1}\left[{}_{RL}D^{\alpha+j-n}_{t,b}f(t)\cdot {}_{RL}D^{n-1-j}_{t,b}g(t)\right]_0^b \tag{7}$$

where ${}_{RL}D^{n-1-j}_{t,b}g(t) = (-1)^{n-1-j}g^{(n-1-j)}(t)$ and $n-1 < \alpha < n \in Z^+$.

In particular, when $0 < \alpha < 1$, if $g(0) = g(b) = 0$ or ${}_{RL}D^{\alpha-1}_{t,b}f(t)|_{t=0} = 0$ holds, then we have

$$\int_0^b f(t)\cdot {}_{C}D^{\alpha}_{0,t}g(t)\mathrm{d}t = \int_0^b g(t)\cdot {}_{RL}D^{\alpha}_{t,b}f(t)\mathrm{d}t \tag{8}$$

when $1 < \alpha < 2$, the following relationship:

$$\int_0^b f(t)\cdot {}_{C}D^{\alpha}_{0,t}g(t)\mathrm{d}t = \int_0^b g(t)\cdot {}_{RL}D^{\alpha}_{t,b}f(t)\mathrm{d}t \tag{9}$$

holds for $g(0) = g(b) = g'(0) = g'(b) = 0$ or $g'(0) = 0$, $g(t) = o(b-t)^{\alpha-1}$ as $t \to b^-$.

Similarly, one has the following property.

Property 2. For suitably smooth functions $f(t)$ and $g(t)$, there holds

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$$\int_0^b f(t)\cdot {}_{RL}D_{0,t}^{\alpha}g(t)\mathrm{d}t = \int_0^b g(t)\cdot {}_{C}D_{t,b}^{\alpha}f(t)\mathrm{d}t - \sum_{j=0}^{n-1}(-1)^{n+j}[{}_{RL}D_{0,t}^{\alpha+j-n}g(t)\cdot f^{(n-1-j)}(t)]_0^b \tag{10}$$

where $n-1<\alpha<n\in Z^+$.

Next, we introduce definitions and properties of Mittag-Leffler function which plays an important role in FODSs.

Definition 4. *The single-parameter Mittag-Leffler function is defined by* [11,14]

$$E_\alpha(x)=\sum_{k=0}^{\infty}\frac{x^k}{\Gamma(\alpha k+1)},\quad \alpha>0$$

Definition 5. *The two-parameter Mittag-Leffler function is defined as* [11,14]

$$E_{\alpha,\beta}(x)=\sum_{k=0}^{\infty}\frac{x^k}{\Gamma(\alpha k+\beta)},\quad \alpha>0,\ \beta>0$$

Obviously,

$$E_1(x)=e^x,\quad E_{\alpha,1}(x)=E_\alpha(x)$$

On the other hand, we introduce several important definitions and results for Lyapunov–Schmidt reduction [15,16].

Definition 6. *Let X and Y be Banach spaces. A bounded linear operator $L:X\to Y$ is called Fredholm if the following three conditions hold*:

(i) *The kernel of L, ker(L) is finite dimensional in X.*
(ii) *The range of L, range(L) is a closed subspace of Y.*
(iii) *The co-kernel of L, coker(L) is also finite dimensional in Y.*

Definition 7. *For Fredholm operator L, the index of L is defined as*

$$i(L)=\dim(\ker(L))-\dim(\mathrm{coker}(L))$$

In the sequel, we introduce the space decomposition principle.

Lemma 1. *If L is Fredholm, then there exist a closed subspace M of X and a closed subspace N of Y, respectively, such that*

$$X=\ker(L)\oplus M,\quad Y=\mathrm{range}(L)\oplus N$$

Furthermore, if the index of L is zero, then one has the orthogonal complement decomposition, namely, $M=(\ker(L))^{\perp}$ and $N=(\mathrm{range}(L))^{\perp}$, where $S^{\perp}=\{u\in X|\langle u,v\rangle=0, \text{for all } v\in S\}$ and $S\subseteq X$. The standard inner product is defined by $\langle u,v\rangle=\int_0^b u(t)v(t)\mathrm{d}t$.

In order to facilitate the explanation of linear operator L which is Fredholm, we present a useful lemma as follows [17].

Lemma 2. *A bounded operator $L:X\to Y$ is Fredholm, if and only if range(L) is closed, and both ker(L) and ker(L^*) are finite dimensional where L^* is the adjoint operator of L.*

3 Lyapunov–Schmidt Reduction

We first consider the case with $0<\alpha<1$.

The Caputo-type FODS with order $\alpha\in(0,1)$ is in the following form:

$${}_{C}D_{0,t}^{\alpha}u(t)=\lambda u(t)+F_1(u) \tag{11}$$

where $F_1(u)=O(u^2)$ as $u\to 0$.

Denote a mapping, $\Phi_1:X_1\times R\to Y$

$$\Phi_1(u,\lambda)\equiv {}_{C}D_{0,t}^{\alpha}u(t)-\lambda u(t)-F_1(u)=0 \tag{12}$$

Choose $X_1=\{u\in C^1[0,b]|\ {}_{RL}D_{t,b}^{\alpha-1}u(t)|_{t=0}=0\}$ and $Y=C_0^0[0,b]=\{h(x)\in C^0[0,b],\ h(0)=0\}$ [11].

Now, we have the following conclusion.

Result 1. For $L_1:X_1\to Y$, X_1 and Y are endowed with the general inner product, then

$$L_1\equiv \mathrm{d}\Phi_1(u,\lambda)/\mathrm{d}u|_{(0,\lambda)}={}_{C}D_{0,t}^{\alpha}-\lambda$$

is Fredholm.

We give a brief explanation. Utilizing Corollary 2.3 of Chap. 2 in Ref. [11], L_1 is bounded. In addition, ker $(L_1)=\{v\in X_1|L_1v=0\}$ implies that $v(t)$ is the solution of the following equation:

$${}_{C}D_{0,t}^{\alpha}v(t)-\lambda v(t)=0 \tag{13}$$

It immediately follows that:

$$v(t)=v(0)E_\alpha(\lambda t^\alpha) \tag{14}$$

So, we get ker $(L_1)=$ span$\{e\}=$ span$\{E_\alpha(\lambda t^\alpha)\}$, therefore, dim (ker $(L_1))=1$.

Next, determine the space ker(L_1^*).

Due to Eq. (8) and definitions of spaces X_1 and Y, it follows that $\langle L_1v_1,v_2\rangle=\langle v_1,L_1^*v_2\rangle$, where $v_1,v_2\in X_1$ and L_1^* is the adjoint operator of L_1. Therefore,

$$\begin{aligned}\langle L_1v_1,v_2\rangle&=\int_0^b({}_{C}D_{0,t}^{\alpha}v_1(t)-\lambda v_1(t))v_2(t)\mathrm{d}t\\&=\int_0^b v_1(t)({}_{RL}D_{t,b}^{\alpha}v_2(t)-\lambda v_2(t))\mathrm{d}t\end{aligned} \tag{15}$$

From Eq. (15), we obtain that $L_1^*={}_{RL}D_{t,b}^{\alpha}-\lambda$. From

$${}_{RL}D_{t,b}^{\alpha}e^*(t)-\lambda e^*(t)=0 \tag{16}$$

one can get $e^*=(b-t)^{\alpha-1}E_{\alpha,\alpha}(\lambda(b-t)^\alpha)$. In the following, we show that:

$$\begin{aligned}&{}_{RL}D_{t,b}^{\alpha}\left((b-t)^{\alpha-1}E_{\alpha,\alpha}\left(\lambda(b-t)^\alpha\right)\right)\\&=\frac{-1}{\Gamma(1-\alpha)}\frac{d\cdot}{dt}\sum_{k=0}^{\infty}\frac{\lambda^k}{\Gamma(\alpha k+\alpha)}\int_t^b(\tau-t)^{-\alpha}(b-\tau)^{\alpha k+\alpha-1}\mathrm{d}\tau\end{aligned} \tag{17}$$

By simple computations

$$\begin{aligned}\int_t^b(\tau-t)^{-\alpha}(b-\tau)^{\alpha k+\alpha-1}\mathrm{d}\tau&=(b-t)^{\alpha k}\int_0^1(1-\xi)^{\alpha k+\alpha-1}(\xi)^{-\alpha}\mathrm{d}\tau\\&=(b-t)^{\alpha k}B(-\alpha+1,\ \alpha k+\alpha)\\&=(b-t)^{\alpha k}\frac{\Gamma(-\alpha+1)\Gamma(\alpha k+\alpha)}{\Gamma(\alpha k+1)}\end{aligned} \tag{18}$$

where $B(z,w)$ is the Beta function.

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Substituting Eq. (18) into Eq. (17) gives

$$
\begin{aligned}
&{}_{RL}D^{\alpha}_{t,b}\Big((b-t)^{\alpha-1}E_{\alpha,\alpha}\big(\lambda(b-t)^{\alpha}\big)\Big)\\
&\quad=-\frac{d}{dt}\sum_{k=0}^{\infty}\frac{\lambda^k(b-t)^{\alpha k}}{\Gamma(\alpha k+1)}\\
&\quad=\frac{\lambda(b-t)^{\alpha-1}}{\Gamma(\alpha)}+\frac{\lambda^2(b-t)^{2\alpha-1}}{\Gamma(2\alpha)}+\frac{\lambda^3(b-t)^{3\alpha-1}}{\Gamma(3\alpha)}+\cdots\\
&\quad=\lambda\Big((b-t)^{\alpha-1}E_{\alpha,\alpha}\big(\lambda(b-t)^{\alpha}\big)\Big)
\end{aligned}
\tag{19}
$$

which shows $\ker(L_1^*)=\mathrm{span}\{e^*\}$. So, dim (ker $(L_1^*))=1$, by Lemma 2, L_1 is Fredholm with index zero.

Next, reducing system (11) into a simple one via the above Lyapunov–Schmidt method, which is divided into five steps.

Step 1 (*Splitting Spaces*). Because dim (ker $(L_1))=1$, and L_1 is Fredholm with index zero, we have the following decomposition:

$$X_1=\ker(L_1)\oplus M,\quad Y=\mathrm{range}(L_1)\oplus N \tag{20}$$

where $M=(\ker(L_1))^{\perp}$, $N=(\mathrm{range}(L_1))^{\perp}$.

Step 2 (*Projection Mapping*). Define $E: Y\to\mathrm{range}(L_1)$ and supplementary projection $I-E: Y\to N$, here I is the identity operator. It follows from Eq. (12) that:

$$\begin{cases}E\Phi_1(u,\lambda)=0\\(I-E)\Phi_1(u,\lambda)=0\end{cases} \tag{21}$$

Step 3 (*Implicit Function Theorem*). For all $u\in X_1$, we always have $u=v+w$, where $v\in\ker(L_1)$, $w\in M$. So, Eq. (21) can be rewritten as

$$\begin{cases}E\Phi_1(v+w,\lambda)=0\\(I-E)\Phi_1(v+w,\lambda)=0\end{cases} \tag{22}$$

Let $\phi(v,w,\lambda):=E\Phi_1(v+w,\lambda)=0$, $\phi(0,0,\lambda)=0$. Hence,

$$
\begin{aligned}
\left.\frac{\mathrm{d}\phi(v,w,\lambda)}{\mathrm{d}w}\right|_{(0,0,\lambda)}&=\left.\frac{\mathrm{d}(E\Phi_1(v+w,\lambda))}{\mathrm{d}w}\right|_{(0,\lambda)}\\
&=E\frac{\mathrm{d}\Phi_1}{\mathrm{d}u}\cdot\left.\frac{\partial u}{\partial w}\right|_{(0,\lambda)}=EL_1=L_1
\end{aligned}
$$

where L_1 is restricted on M. By means of the implicit function theorem, there exists a unique solution $w=f(v,\lambda)$ which satisfies Eq. (22). In general, $w=f(v,\lambda)$ cannot be analytically expressed.

Step 4 (*Substitution*). Let $\Psi_1:\ker(L_1)\times R\to N$

$$\Psi_1(v,\lambda)\equiv(I-E)\Phi(v+f(v,\lambda),\lambda)=0 \tag{23}$$

The above equation is finite dimensional.

Step 5 (*Reduction Equation*). Due to Fredholm alternative [16]

$$(\mathrm{range}(L_1))^{\perp}=\ker(L_1^*) \tag{24}$$

one has

$$
\begin{aligned}
g(x,\lambda)&:=\langle e^*,\Psi_1(v,\lambda)\rangle\\
&=\langle e^*,(I-E)\Phi_1(v,\lambda)\rangle\\
&=\langle e^*,\Phi_1(v,\lambda)\rangle-\langle e^*,E\Phi_1(v,\lambda)\rangle\\
&=\langle e^*,\Phi_1(xe+f(xe,\lambda),\lambda)\rangle=0
\end{aligned}
\tag{25}
$$

where $e=E_{\alpha}(\lambda t^{\alpha})$, $e^*=(b-t)^{\alpha-1}E_{\alpha,\alpha}(\lambda(b-t)^{\alpha})$, and $x\in R$, $e\in\ker(L_1)$, $e^*\in\ker(L_1^*)$.

$g(x,\lambda)=0$ is called the reduction equation. Then, the local dynamics of the original system (11) near $(0,\lambda)$ can be got from its reduced equation $g(x,\lambda)=0$. If λ is a bifurcation parameter, then $g(x,\lambda)=0$ is the corresponding bifurcation equation (8).

Next, we study the following equation with order $\alpha\in(1,2)$:

$${}_{C}D^{\alpha}_{0,t}u(t)=\lambda u(t)+F_2(u) \tag{26}$$

where $t\in[0,b]$, $F_2(u)=O(u^2)$ as $u\to 0$.

Set $X_2=\{u\in C^2[0,b]|\ u'(0)=0,u(t)=o(b-t)^{\alpha-1}\}$ and $Y=C_0^0[0,b]=\{h(x)\in C^0[0,b],\ h(0)=0\}$ [11].

Denote also a mapping: $\Phi_2: X_2\times R\to Y$

$$\Phi_2(u,\lambda)\equiv{}_{C}D^{\alpha}_{0,t}u(t)-\lambda u(t)-F_2(u)=0 \tag{27}$$

Similarly, we can get the linearization of Φ_2 near $(0,\lambda)$, namely, $L_2\equiv(\mathrm{d}\Phi_2(u,\lambda)/\mathrm{d}u)|_{(0,\lambda)}$.

Result 2. For $L_2: X_2\to Y$, X_2 and Y are equipped with the standard inner product, then

$$L_2=\left.\frac{\mathrm{d}\Phi_2(u,\lambda)}{\mathrm{d}u}\right|_{(0,\lambda)}={}_{C}D^{\alpha}_{0,t}-\lambda$$

is Fredholm.

The proof is almost the same as that of case $\alpha\in(0,1)$, so is omitted here.

It is easy to show that

$$\ker(L_2)=\mathrm{span}\{e_1,e_2\}$$

where $e_1=E_{\alpha}(\lambda t^{\alpha})$ and $e_2=tE_{\alpha,2}(\lambda t^{\alpha})$. From the Fredholm alternative, we obtain that coker $(L_2)=$ ker (L_2^*), in which L_2^* is the adjoint operator associated with L_2, satisfying $\langle L_2v_1,v_2\rangle=\langle v_1,L_2^*v_2\rangle$, where $v_1,v_2\in X_2$.

It is evident that $L_2^*={}_{RL}D^{\alpha}_{t,b}-\lambda$, and

$$\ker(L_2^*)=\mathrm{span}\{e_1^*,e_2^*\}$$

in which $e_1^*=(b-t)^{\alpha-1}E_{\alpha,\alpha}(\lambda(b-t)^{\alpha})$, and $e_2^*=(b-t)^{\alpha-2}E_{\alpha,\alpha-1}(\lambda(b-t)^{\alpha})$.

System (26) can be changed into a reduced system also by five steps.

Step 1 (*Splitting Spaces*). Based on the above discussion, we clearly know that L_2 is Fredholm with index zero and dim (ker $(L_2))=2$. Naturally, we get the following orthogonal complements:

$$X_2=\ker(L_2)\oplus M_2,\quad Y=\mathrm{range}(L_2)\oplus N_2 \tag{28}$$

where $M_2=(\ker(L_2))^{\perp}$, $N_2=(\mathrm{range}(L_2))^{\perp}$.

Step 2 (*Projection Mapping*). Define projective operators, $E_2: Y\to\mathrm{range}(L_2)$ and $I-E_2: Y\to N_2$, where I is the identity operator. So, Eq. (27) is

$$\begin{cases}E_2\Phi_2(u,\lambda)=0\\(I-E_2)\Phi_2(u,\lambda)=0\end{cases} \tag{29}$$

Step 3 (*Implicit Function Theorem*). By Eq. (28), for arbitrary $u\in X_2$, we have $u=v+w$, where $v\in\ker(L_2)$, $w\in M_2$. So, Eq. (29) can be rewritten as

$$\begin{cases}E_2\Phi_2(v+w,\lambda)=0\\(I-E_2)\Phi_2(v+w,\lambda)=0\end{cases} \tag{30}$$

Due to the implicit function theorem, from the first equation of Eq. (30) there exists a unique solution $w=g(v,\lambda)$. The exact expression of w is often difficult to obtain.

Step 4 (*Substitution*). Define a mapping $\Psi_2:\ker(L_2)\times R\to N_2$

$$\Psi_2(v,\lambda)\equiv(I-E_2)\Phi_2(v+g(v,\lambda),\lambda)=0 \tag{31}$$

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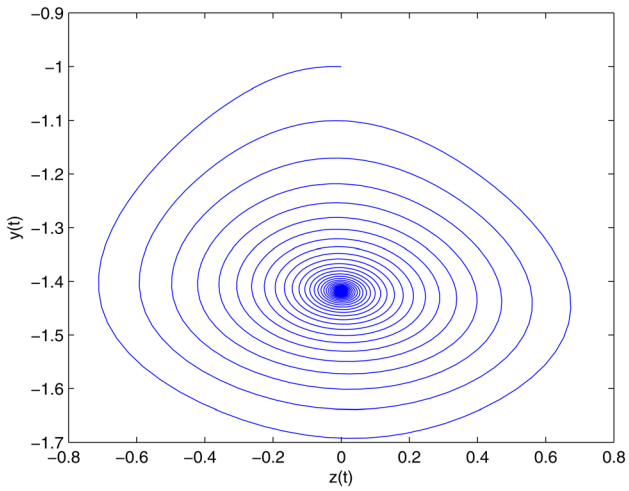

Fig. 1 Phase diagram of system (36) when $\lambda = -0.2$, $\alpha = 1.9$, $x(0) = 0$, and $y(0) = -1$

Step 5 (*Reduction Equation*). Using Fredholm alternative, we can select

$$(\mathrm{range}(L_2))^{\perp} = \ker(L_2^*) \tag{32}$$

So after tedious calculations, we can obtain the following reduction equations of the original system (26):

$$\begin{aligned} &h_1(x_1, x_2, \lambda) = \\ &\langle e_1^*, \Phi_2(x_1 e_1 + x_2 e_2 + g(x_1 e_1 + x_2 e_2, \lambda), \lambda)\rangle = 0 \end{aligned} \tag{33}$$

and

$$\begin{aligned} &h_2(x_1, x_2, \lambda) = \\ &\langle e_2^*, \Phi_2(x_1 e_1 + x_2 e_2 + g(x_1 e_1 + x_2 e_2, \lambda), \lambda)\rangle = 0 \end{aligned} \tag{34}$$

where $e_1 = E_\alpha(\lambda t^\alpha)$, $e_2 = tE_{\alpha,2}(\lambda t^\alpha)$, $e_1^* = (b-t)^{\alpha-1} E_{\alpha,\alpha}(\lambda(b-t)^\alpha)$, $e_2^* = (b-t)^{\alpha-2} E_{\alpha,\alpha-1}(\lambda(b-t)^\alpha)$, and $(x_1, x_2) \in R^2$, $e_1, e_2 \in \ker(L_2)$, $e_1^*, e_2^* \in \ker(L_2^*)$.

$h_1(x_1, x_2, \lambda) = 0$ and $h_2(x_1, x_2, \lambda) = 0$ are called the reduction equations of system (26). Then, the local dynamics of the original system (26) near $(0, \lambda)$ can be got from its reduced equations $h_1(x_1, x_2, \lambda) = h_2(x_1, x_2, \lambda) = 0$. If λ is a bifurcation parameter, then they are the corresponding bifurcation equations (8).

Remark 1. (1) The dynamics of the original systems (11) and (26) can be understood from their reduction systems (25), (33), and (34), respectively.

(2) Generally speaking, it is often difficult even impossible to get the exact solution of the reduction equation. So, it turns to find its approximate solution.

In the sequel, we give an example as follows.

Example 1. We consider the nonlinear FODS with Caputo derivative with derivative order $\alpha \in (1, 2)$

$${}_C D_{0,t}^{\alpha} y(t) = \lambda y(t) + \sin(y^3(t)) \tag{35}$$

where $y(t) \in X_2$, which is defined as in case with $1 < \alpha < 2$. System (35) can be transformed into the following system:

$$\begin{cases} y'(t) = z(t) \\ {}_C D_{0,t}^{\alpha-1} z(t) = \lambda y(t) + \sin(y^3(t)) \end{cases} \tag{36}$$

Its phase diagram is presented in Fig. 1 with $\lambda = -0.2$, $\alpha = 1.9$, $x(0) = 0$, and $y(0) = -1$. By virtue of the above reduction analysis, we can obtain the reduced equations for system (35) as below

$$\begin{aligned} &h_1(x_1, x_2, \lambda) \\ &= \int_0^b (b-\tau)^{\alpha-1} E_{\alpha,\alpha}(\lambda(b-\tau)^\alpha) \cdot [{}_C D_{0,\tau}^{\alpha} g(x_1 E_\alpha(\lambda\tau^\alpha) \\ &\quad + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha), \lambda) - \lambda g(x_1 E_\alpha(\lambda\tau^\alpha) + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha), \lambda) \\ &\quad - \sin(x_1 E_\alpha(\lambda\tau^\alpha) + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha) + g((x_1 E_\alpha(\lambda\tau^\alpha) \\ &\quad + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha), \lambda)))^3] \mathrm{d}\tau = 0 \end{aligned} \tag{37}$$

and

$$\begin{aligned} &h_2(x_1, x_2, \lambda) \\ &= \int_0^b (b-\tau)^{\alpha-2} E_{\alpha,\alpha-1}(\lambda(b-\tau)^\alpha) \cdot [{}_C D_{0,\tau}^{\alpha} g(x_1 E_\alpha(\lambda\tau^\alpha) \\ &\quad + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha), \lambda) - \lambda g(x_1 E_\alpha(\lambda\tau^\alpha) + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha), \lambda) \\ &\quad - \sin(x_1 E_\alpha(\lambda\tau^\alpha) + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha) + g((x_1 E_\alpha(\lambda\tau^\alpha) \\ &\quad + x_2 \tau E_{\alpha,2}(\lambda\tau^\alpha), \lambda)))^3] \mathrm{d}\tau = 0 \end{aligned} \tag{38}$$

Next, we compute the derivatives of these reduction equations at the origin. Before starting the calculations, let $x = (x_1, x_2)$, $e = (e_1, e_2)^\top$, and $e^* = (e_1^*, e_2^*)^\top$. And, we introduce an invariant notation for higher-order Gâteaux derivatives of mapping $\Phi_2 \equiv {}_C D_{0,t}^{\alpha} y(t) - \lambda y(t) - \sin(y^3(t))$ [16]

$$D^k \Phi_2(u, \lambda)(v_1, \ldots, v_k) = \frac{\partial^k}{\partial t_1 \cdots \partial t_k} \Phi_2\left(u + \sum_{i=1}^{k} t_i v_i, \lambda\right)\Bigg|_{t_i=0} \tag{39}$$

After tedious calculations and combining the results of Ref. [16], we know that

(a) $h_1(0,0) = h_{1x}(0,0) = h_{1xx}(0,0) = h_{1\lambda}(0,0) = 0$,
(b) $h_{1xxx} = \langle e^*, D^3\Phi_2(e,e,e)\rangle = \langle e^*, -6e^3\rangle < 0$, and
(c) $h_{1\lambda x} = \langle e^*, D\Phi_{2\lambda}(e)\rangle = \langle e^*, -e\rangle < 0$.

So utilizing the results on the bifurcation recognition problem [16,18], we know that $h_1 = h_2 = 0$ are locally equivalent to $x^3 + \lambda x = 0$ near the origin. The corresponding phase diagram is displayed in Ref. [18].

4 Conclusions

In this paper, we extend the Lyapunov–Schmidt reduction for ODEs to FODSs. The local dynamical behaviors can be known from their reduction systems. We mainly study Caputo-type FODSs with fractional order $\alpha \in (0, 1)$ and $\alpha \in (1, 2)$ and get their reduction systems. Finally, an example is presented to illustrate the theoretical analysis which coincides with the derived method.

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